Connected Components in an Undirected Graph

Problem statement

Given a simple undirected graph on vertices 1 through N, determine its number of connected components. Isolated vertices count separately.

Input format

First line: N M. The next M lines each contain an undirected edge U V. Endpoints may be in either order.

Output format

Print one integer: the number of connected components.

Constraints

- 1 <= N <= 120
- 0 <= M <= N*(N-1)/2
- 1 <= U, V <= N; U != V
- No duplicate undirected edges

Sample input and output

Sample input	Expected output
5 3 1 2 2 3 4 5	2

Important boundary conditions

All edges are undirected. A vertex with no edges still forms a component.

Implementation expectations

A matrix-based DFS/BFS is O(N^2). An adjacency list or DSU may be more efficient.

Submission requirements

Submit exactly one .c file. Use stdin and stdout without interactive prompts. Compile with GCC in C11 mode. Avoid undefined behavior, release dynamically allocated memory, and print precisely the required output. Follow the stated input constraints.

Teacher resources (not part of student handout)

A reference oracle, five sample student submissions, assignment-specific rules.json, and 12 verified tests are supplied separately in the instructor test pack. Do not distribute the reference oracle or deliberately faulty variants as student solutions.